

ENGG1320 GROUP 5 REPORT

One by One - Physical Monitoring Productivity App

Date: 3 December 2024

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Programme Code

Facial Detection

<https://drive.google.com/file/d/1qiKhbeySsOczUIJXDnYvRjN0T-bNvloJ/view?usp=sharing>

Distance Detection

<https://drive.google.com/file/d/1mgQzdmv-0nUjHxP0aP65eW5f0HuTffLH/view?usp=sharing>

Chat Analysis

https://drive.google.com/file/d/12cZ-tGxLX_VwcoO4v7DU0MCuuytcijQ1/view?usp=sharing

Software Monitoring

<https://drive.google.com/file/d/1TlhLML-cQ166FeOznakke62refQ3FYzS/view?usp=sharing>

Project Idea

Written by: Zhifang Li

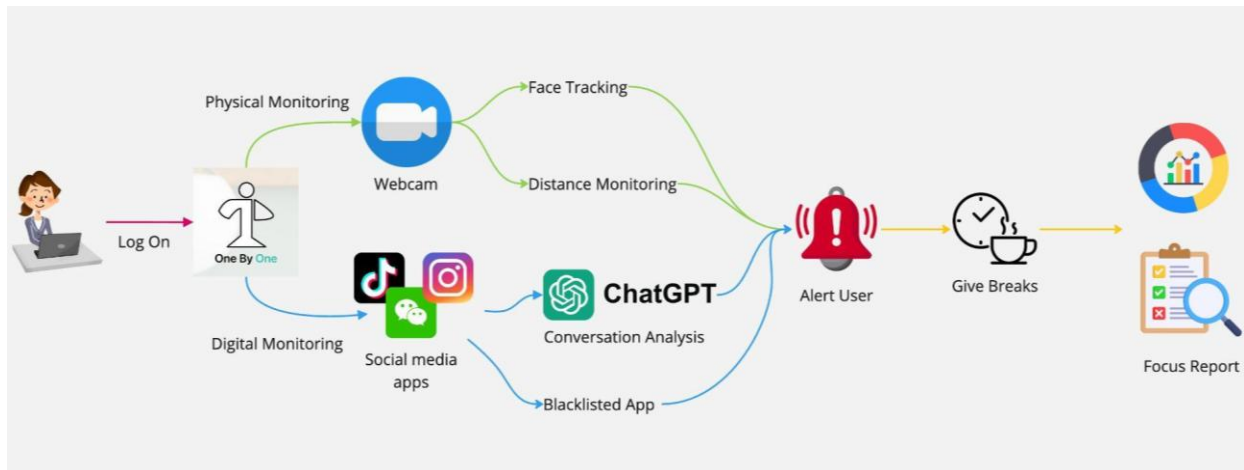


The productivity market is currently experiencing a significant upward trend. Evidenced by the sales of books such as Atomic Habits by James Clear and Deep Work by Cal Newport, as well as the multitude of focus apps and content creators available online, people nowadays no longer crave new entertainment but rather to create peace of mind in an increasingly noisy era.

Hence, our team came up with this idea: a productivity app that can monitor your physical focus. We noticed that with popular apps such as Freedom and Forest, we could stop getting distracted by social media and YouTube, but there was still nothing stopping us from just grabbing a comic book off the shelf. Behavioural therapy shows how important impulse control is - even a minor action can lead you to an entire spiral. Thus, we wanted to create a product that would help train its users to become focused for longer so they would have more time afterwards to do the things they truly enjoy. The name of this product is One by One, as we hope to help our users focus on tasks one at a time.

The purpose of this app is to help the user become more disciplined over time. While we do not want users to become reliant on us, we do hope to help them when they're first starting out by gently alerting them if they're distracted. Ours will have two main functions: digital monitoring and physical monitoring. The former will check whether the apps and chats you open are work-related,

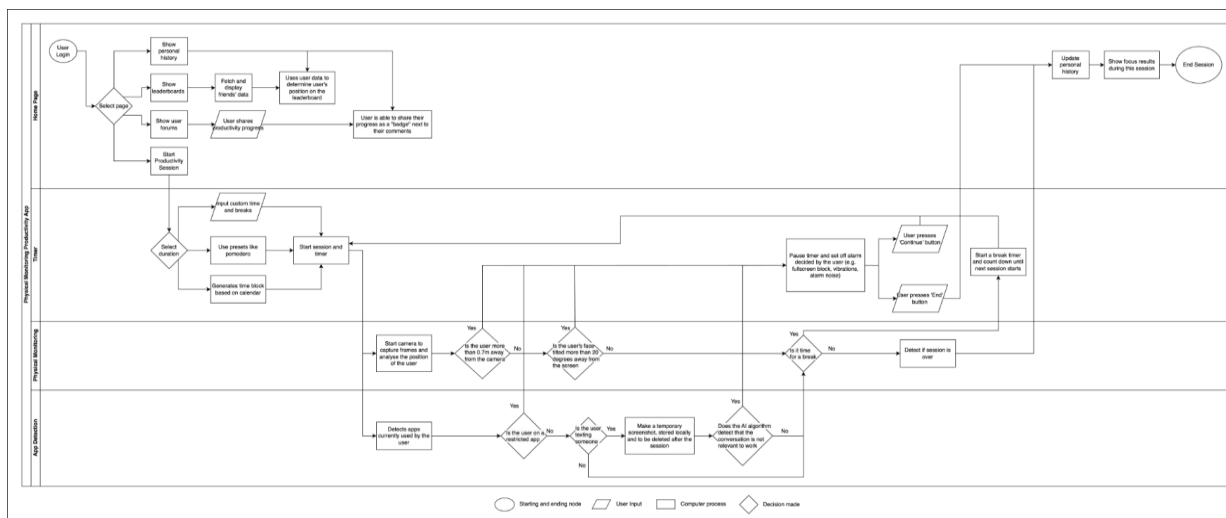
while the latter will check if you're looking at the screen or not. Going beyond a set parameter for both will cause the app to alert you for being distracted. When each session has ended, the user will be sent a short report of all the distractions they had during this period.



(For better resolution, view here:

<https://drive.google.com/file/d/10nf7IERPbVrPZL3cFGXGcK3M2M28Jbx1/view?usp=sharing>)

Additional features of this app include a built-in leaderboard system and a productivity forum for users to discuss their progress and techniques with each other. Both of these features are relatively simple ways to motivate the average user and help the more ambitious users better themselves faster along their productivity journey.



(Due to resolution constraints, the swimlane diagram cannot be fitted to an appropriate scale here, viewable in full size through this link:

https://drive.google.com/file/d/1vIZ3AGzN89-zh9AfU8BU4H2rP0e_7xhK/view?usp=sharing)

With this project, we aim to first become a self-sustaining business and to distribute this software worldwide, building an international community of like-minded, focused people.

Core Contents

Facial Tracking

Core Contributors: ZhangJiachang

Overview

This component aims to detect whether the user is distracted based on the angle at which the user's face is tilted. If the absolute value of the angle is above 10, the system will display an alert message that warns the user about their behaviour. The system will also display a message when the user is detected as not distracted.

The code for this component references the following repositories from GitHub:

- <https://github.com/yinguobing/head-pose-estimation>
- <https://github.com/Jacob-Mellichamp/DistractorDetector>

The full code for this project can be viewed here:

<https://drive.google.com/file/d/1qiKhbeySsOczUIJXDnYvRjN0T-bNvloJ/view?usp=sharing>

Facial detection modules

This module uses packages such as OpenCV to initialise a face detector class that preprocesses the inputted images (such as converting colours, expanding the image dimensions and transposing the image) and outputs a bounding box surrounding the user's face. This module provides further data for the facial landmark modules as it first locates the coordinates of the user's face. The module relies on a pre-trained model from OpenCV.

```
class FaceDetector:
    """Detect human face from image"""

    def __init__(self,
                 dnn_proto_text='models/deploy.prototxt',
                 dnn_model='models/res10_300x300_ssd_iter_140000.caffemodel'):
        """Initialization"""
        self.face_net = cv2.dnn.readNetFromCaffe(dnn_proto_text, dnn_model)
        self.detection_result = None

    def get_faceboxes(self, image, threshold=0.5):
        """
        Get the bounding box of faces in image using dnn.
        """
        rows, cols, _ = image.shape

        confidences = []
        faceboxes = []

        self.face_net.setInput(cv2.dnn.blobFromImage(
            image, 1.0, (300, 300), (104.0, 177.0, 123.0), False, False))
        detections = self.face_net.forward()

        for result in detections[0, 0, :, :]:
            confidence = result[2]
            if confidence > threshold:
                x_left_bottom = int(result[3] * cols)
                y_left_bottom = int(result[4] * rows)
                x_right_top = int(result[5] * cols)
                y_right_top = int(result[6] * rows)
                confidences.append(confidence)
                faceboxes.append(
                    [x_left_bottom, y_left_bottom, x_right_top, y_right_top])

        self.detection_result = [faceboxes, confidences]

        return confidences, faceboxes
```

Mark Detection Modules

The mark detection modules run based on the pre-trained deep learning CNN (Convolutional Neural Network) model that detects facial landmarks from the face, extracts the face from the images and plots key points on the face. More specifically, the code initialises a MarkDetector with an init function, which reads the existing TensorFlow model and sets the input size to 128. Then, it calls the preprocess function to resize each inputted image and convert them to RGB format. The programme extracts the faces from each image using the coordinates from the faceDetector modules and creates visualisations based on these coordinates.

```
class MarkDetector:
    """Facial landmark detector by Convolutional Neural Network"""

    def __init__(self, model_file):
        """Initialize a mark detector.

        Args:
            model_file (str): ONNX model path.
        """
        assert os.path.exists(model_file), f"File not found: {model_file}"
        self._input_size = 128
        self.model = ort.InferenceSession(
            model_file, providers=["CUDAExecutionProvider", "CPUExecutionProvider"])

    def _preprocess(self, bgrs):
        """Preprocess the inputs to meet the model's needs.

        Args:
            bgrs (np.ndarray): a list of input images in BGR format.

        Returns:
            tf.Tensor: a tensor
        """
        rgbs = []
        for img in bgrs:
            img = cv2.resize(img, (self._input_size, self._input_size))
            img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
            rgbs.append(img)

        return rgbs
```

```
def detect(self, images):
    """Detect facial marks from an face image.

    Args:
        images: a list of face images.

    Returns:
        marks: the facial marks as a numpy array of shape [Batch, 68*2].
    """
    inputs = self._preprocess(images)
    marks = self.model.run(["dense_1"], {"image_input": inputs})
    return np.array(marks)

def extract_cnn_facebox(self, image):
    """Extract face area from image."""
    raw_boxes = self.face_detector.get_faceboxes(
        image=image, threshold=0.5)
    a = []
    for box in raw_boxes:
        offset_y = int(abs((box[3] - box[1]) * 0.1))
        box_moved = self.move_box(box, [0, offset_y])

        # Make box square.
        facebox = self.get_square_box(box_moved)

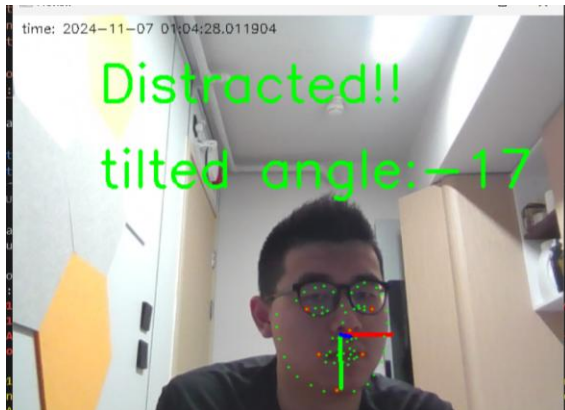
        if self.box_in_image(facebox, image):
            a.append(facebox)

    return a

def visualize(self, image, marks, color=(255, 255, 255)):
    """Draw mark points on image"""
    for mark in marks:
        cv2.circle(image, (int(mark[0]), int(
            mark[1])), 1, color, -1, cv2.LINE_AA)
```

Main Modules

The main module referenced parameters such as camera matrix and distortion coefficient provided by the author Jacob Mellichamp, which are two key parts needed to calculate the tilted angle of the user's nose. The main function imports both the aforementioned facial detection and mark detection modules and opens the camera using `CV2.VideoCapture()`. The tangent angle function is also included to calculate the degree between two nose points. Then, if the absolute value of the angle exceeds 10, the system will output a message that shows the user they are distracted and vice versa.



1. Example of the user interface
2. Code snippet of the main module
3. Code snippet of the main module

```
def calcAngle(p1, p2):
    ang = None
    try:
        m = (p1[1] - p2[1]) / (p1[0] - p2[0])
        ang = int(math.degrees(math.atan(-1/m)))
    except:
        ang = 90
    return ang

def run():
    # Before estimation started, there are some startup works to do.

    # Initialize the video source from webcam or video file.
    video_src = args.cam if args.video is None else args.video
    cap = cv2.VideoCapture(video_src)
    print(f"Video source: {video_src}")

    # Get the frame size. This will be used by the following detectors.
    frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
    frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
    model_points = np.array([
        (0.0, 0.0, 0.0),          # Nose tip
        (0.0, -330.0, -65.0),     # Chin
        (-225.0, 170.0, -135.0),  # Left eye left corner
        (225.0, 170.0, -135.0),   # Right eye right corne
        (-150.0, -150.0, -125.0), # Left Mouth corner
        (150.0, -150.0, -125.0)   # Right mouth corner
    ])

    for p in image_points:
        cv2.circle(frame, (int(p[0]), int(p[1])), 3, (0,0,255), -1)

    p1 = (int(nose_points[0][0]), int(nose_points[0][1]))
    p2 = (int(nose_points[3][0]), int(nose_points[3][1]))

    #cv2.line(frame, p1, p2, (255, 255, 0), 2)

    ang = calcAngle(p1, p2)
    if abs(ang) > 10 :
        cv2.putText(frame, f"tilted angle:{str(ang)}", (100, 200), cv2.FONT_HERSHEY_SIMPLEX, 2, (0, 255, 0), 3)
        cv2.putText(frame, str("Distracted!!"), (100, 100), cv2.FONT_HERSHEY_SIMPLEX, 2, (0, 255, 0), 3)
    else:
        cv2.putText(frame, str("NOT Distracted!!"), (100, 100), cv2.FONT_HERSHEY_SIMPLEX, 2, (0, 255, 0), 3)
        cv2.putText(frame, f"tilted angle:{str(ang)}", (100, 200), cv2.FONT_HERSHEY_SIMPLEX, 2, (0, 255, 0), 3)
```

Distance Detection

Core Contributors: Chau Wai Yee

Overview

This component, written in Python, is supported by the YOLOv8 object detection model and OpenCV library. With the help of a webcam, it employs computer vision techniques to process object detection and measures the real-time distance between the user and their webcam. If the app detects that the distance exceeds 0.7m, it will send an alert. This distance detection is referenced from the code written by Takagi on GitHub (2024).

Requirement

Python 3.8 or later, OpenCV, and Ultralytics YOLO library ought to be installed before running this program. Code for this project can be viewed here:

<https://drive.google.com/file/d/1mgQzdmv-0nUjHxP0aP65eW5f0HuTffLH/view?usp=sharing>

Procedure of distance detection

1. Define initial values, including 1) pre-trained YOLOv8 model, 2) video capture object, 3) frame width, 4) frame height, 5) frame per second, 6) video writer object, 7) centre point in the frame, 8) pixels per meter, 9) text colour, and 10) text background colour:

```
model = YOLO("yolov8s.pt")
cap = cv2.VideoCapture(0)

w, h, fps = (int(cap.get(x)) for x in (cv2.CAP_PROP_FRAME_WIDTH, cv2.CAP_PROP_FRAME_HEIGHT, cv2.CAP_PROP_FPS))
fps = 3.33

out = cv2.VideoWriter('output.avi', cv2.VideoWriter_fourcc(*'MJPG'), fps, (w, h))

center_point = (w // 2, h // 2)
pixel_per_meter = 250

txt_color, txt_background = ((0, 0, 0), (255, 255, 255))
```

2. Inside a while loop, if the frame is unsuccessfully captured, all OpenCV windows will be closed. Otherwise, the YOLO machine learning model is applied to the actual frame captured from the video source (im0) to track the person in im0. Then, we can access the

bounding box coordinates of the upper-left corner and lower-left corner of each tracked object in the first result from the tracking output.

3. Inside the same while loop, unique IDs of tracked objects are converted into a list of integers (track_ids). If there is no 1, which is the ID for the person, in the list, the text “Distracted!” is added to im0 because the user has left, and no person object could be tracked from the model. Otherwise, we use a for loop to find the person object. Once we find the person object, we measure the distance between the computer and the user by dividing the Euclidean distance between the centre point of the frame and the centroid of the bounding box for the person object by the pixel-per-meter ratio. This distance value is then added to im0. If the distance is larger than 0.7 meters (m), the text “Distracted!” will also be added to im0; otherwise, the text “No Distracted!!” is added.

```
while True:
    ret, im0 = cap.read()
    if not ret:
        print("Video frame is empty or video processing has been successfully completed.")
        break

    results = model.track(im0, persist=True)
    boxes = results[0].boxes.xyxy.cpu()
```

Chat Analysis

Core Contributor: Wong Suet Nga

Overview

The Chat Analysis function determines whether or not the user is distracted by analysing the user's chats on screen, judging if it is “work-related” or “non-work-related”. Once the chat content is determined as “non-work-related”, a focus reminder will appear on the user's screen.

Implementation

This program was written in Python 3.8, with imported libraries and engines from Tesseract-OCR, spaCy, Tkinter, and PyAutoGUI. References are made to generated code written by Poe (<https://poe.com/>).

The code for this project is viewable here:

https://drive.google.com/file/d/12cZ-tGxLX_VwcoO4v7DU0MCuuytcijQ1/view?usp=sharing

Step 1: Extract text from user's screen

Under the user's explicit consent, the user's screen will be captured every 15 seconds. Then, text is extracted from the screenshot.

```
def capture_screen():
    screenshot = pyautogui.screenshot()
    screenshot.save("screenshot.png")
    return "screenshot.png"

def extract_text(image_path):
    img = Image.open(image_path)
    text = pytesseract.image_to_string(img)
    return text
```

Step 2: Processing text

The text is processed by spaCy, a natural language processor, to interpret the attitude of people participating in the chat. SpaCy first filters out meaningless words (e.g. is, to, a, for) from meaningful words (such as the subject and object of a sentence) for further analysis.

Then, it compares the filtered text with a predetermined list of work-related phrases. If the text from the user's chat and the work-related phrases are similar (i.e., their similarity value exceeds a set threshold), or if they are an exact match, that word from the user's text is considered work-related. The counter *workcount* is incremented by 1. If *workcount* exceeds a set threshold, the user's chat is considered work-related, and *True* is returned.

```
str_work_related_phrases = ' '.join(work_related_phrases)
nlp_work_related_phrases = nlp[str_work_related_phrases]

for token in doc: # word-by-word
    if token.dep_ in ("nsubj", "dobj", "pobj"): # Subject, object, or obj
        for workphrase in nlp_work_related_phrases:
            if workphrase.similarity(token) > 0.6:
                workcount += 1
            if token.text in work_related_phrases:
                workcount += 1
if workcount > 20:
    return True
else:
    return False
```

Step 3: Remind the user to focus

If the user's chat is judged non-work-related, a message box will pop up on the user's screen to remind them to focus on their work.

```
def show_result(is_work_related):
    if not is_work_related:
        messagebox.showinfo("It appears that your chat is not work-related. Please focus on your work.")
```

Step 4: Looping the program

The main program links each function together and runs them in a loop. It is responsible for starting Tkinter. The user's screen is captured every 5 seconds.

Software Monitoring

Core Contributor: Zeng Xinyu

Overview

The software monitoring function checks if the user is on an entertainment-type application and sends a push notification when detected. This implementation was written in Python 3.8 and used the `ActivityManager` API on Android and the `applicationDidBecomeActive` and `applicationWillResignActive` APIs on iOS. The code for this project is viewable here:

<https://drive.google.com/file/d/1TlhLML-cQ166FeOznakke62refQ3FYzS/view?usp=sharing>

Implementation

Step 1: Obtaining Current Active Application Information

First, we want to obtain information about the application the user is using. This can be achieved through APIs provided by the operating system. On Android, we can use the `ActivityManager` or `UsageStatsManager` APIs to get the usage statistics of applications; `ActivityManager` is a system service that can be used to obtain all running tasks and processes in the system, whereas `UsageStatsManager` provides more detailed usage statistics for applications, though users must manually grant some permissions in the settings. For convenience, we choose to use `ActivityManager`. On iOS, we can make use of the `applicationDidBecomeActive` and `applicationWillResignActive` APIs, which are called when the app enters the foreground or background, allowing us to detect application switches.

```
def main():
    # Create the main Tkinter window
    root = tk.Tk()
    root.title("Chat Monitor")

    # Start the monitoring process
    def start_monitoring():
        while True:
            image_path = capture_screen()
            text = extract_text(image_path)

            result = is_work_related(text)
            show_result(result) # reminder to user

            time.sleep(5) # Adjust the interval as needed

    # Start monitoring in a separate thread to avoid freezing the GUI
    import threading
    monitoring_thread = threading.Thread(target=start_monitoring)
    monitoring_thread.start()

    # Start the Tkinter main loop
    root.mainloop()
```

```

import objc
from Foundation import NSObject, NSNotificationCenter, NSRunLoop, NSDefaultRunLoopMode
from AppKit import NSWorkspace

class AppSwitchDetector(NSObject):
    def init(self):
        self = objc.super(AppSwitchDetector, self).init()
        if self is None: return None
        self.workspace = NSWorkspace.sharedWorkspace()
        self.notification_center = NSNotificationCenter.defaultCenter()
        self.notification_center.addObserver_selector_name_object_(self, 'appDidActivate:', 'NSWorkspaceDidActivateApplicationNotification', None)
        return self

    def appDidActivate_(self, notification):
        app = notification.userInfo().objectForKey_('NSWorkspaceApplicationKey')
        print(f"Active application changed to: {app.localizedName()}")

if __name__ == "__main__":
    detector = AppSwitchDetector.alloc().init()
    run_loop = NSRunLoop.currentRunLoop()
    while True:
        run_loop.runUntilDate_(NSDate.distantFuture())

```

Fig.1 Implementation on iOS

```

#Android--ActivityManager--ADB(Android Debug Bridge)

import os

def get_current_active_app():

    result = os.popen('adb shell dumpsys activity activities | grep mResumedActivity').read()

    if "mResumedActivity" in result:
        package_name = result.split(" ")[-1].strip()
        return package_name
    return None

if __name__ == "__main__":
    current_app = get_current_active_app()
    print(f"Current active application: {current_app}")

```

Fig.2 Implementation on Android

Step 2: Identifying Entertainment Applications

After obtaining the currently active application information, we need to identify whether or not the app is an entertainment-related application. This can either be done manually by creating a list of known entertainment apps, matching the currently active application against this list, or by machine learning classification, which involves training a classifier based on known application features (such as package name, category, description, etc.).

The list approach requires manual real-time updates as the popular entertainment apps are ever-changing, which can be challenging to maintain and is prone to delays. While the machine learning approach is more work-intensive, as it requires data collection, feature extraction, model training, model evaluation, and deployment, it has been chosen here as it is more feasible to maintain for our small team.

```

#Apply Trained Model

import joblib
import numpy as np

model = joblib.load('entertainment_app_classifier.pkl')

def is_entertainment_app(app_features):

    app_features = np.array(app_features).reshape(1, -1)
    prediction = model.predict(app_features)
    return prediction[0] == 1 # Entertainment app

# Sample
sample_app_features = [0.5, 0.3, 0.2, ...]
is_entertainment = is_entertainment_app(sample_app_features)
print(f"Is entertainment app: {is_entertainment}")

```

Step 3: Real Time Monitoring and Notifications

Finally, we implement a mechanism to monitor user activity in real time and send notifications or log events when distraction is detected. First, we set a timer to check the currently active applications at set intervals (e.g. every minute). Then, the programme will listen to system broadcasts and events and trigger when there is an application switch. If an entertainment application is detected, a notification will be sent to remind the user, and a record is made for the final productivity report. With this method, software checks should be fully functional for all of our users.

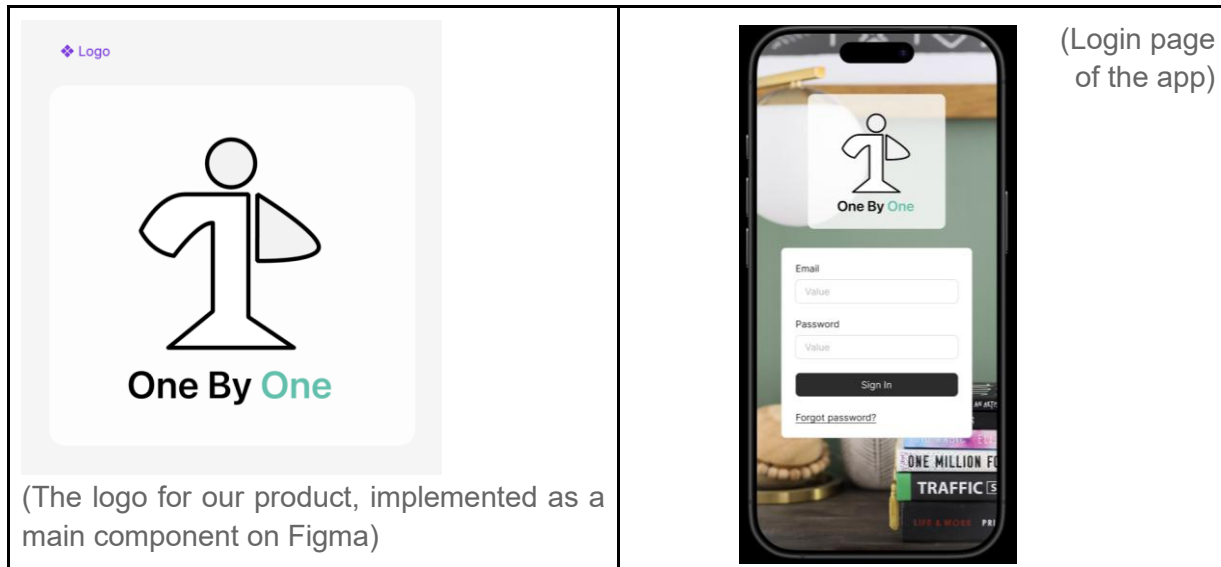
User Interface Design

Core Contributor: Zhifang Li

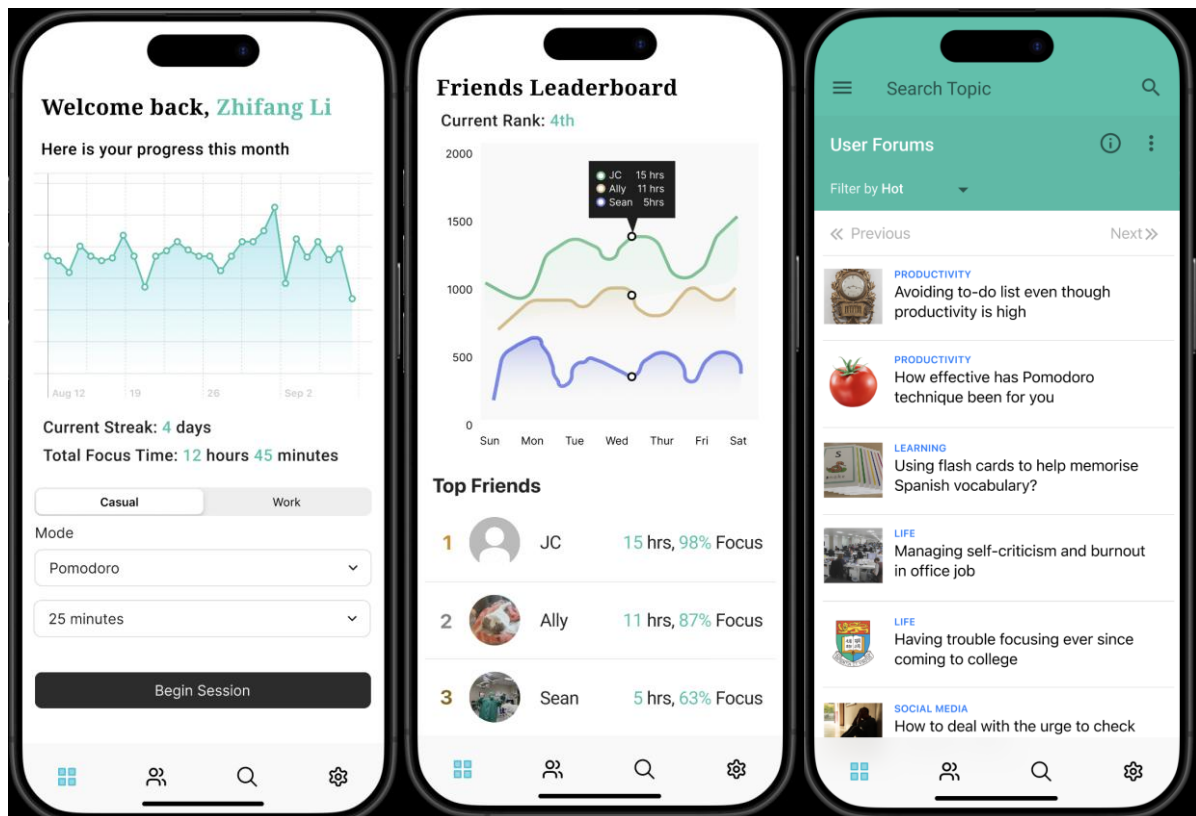
(A mockup of the mobile design can be found at:

<https://www.figma.com/proto/OPDtGFWhXmTA4WRCVFkpZL/1By1-Mobile-UI?node-id=7-104&starting-point-node-id=7%3A104>. Note that due to technical limitations, the prototype is built around the iPhone 16 Pro screen size and may look incorrect if the scale is changed)

For this project, we have decided to create both mobile and web interfaces for our app, as we want the app to be as accessible and ensure there is low friction between the user and our services. Additionally, having the app on both platforms would be convenient as both phones and laptops typically have front-facing cameras, meaning there would need to be minimal setup before our services can be used. What is demonstrated in this section is a prototype of our mobile app, which was made using Figma. As stated before, the name of our product is One by One, as we hope that our users will become focused on one task at a time.



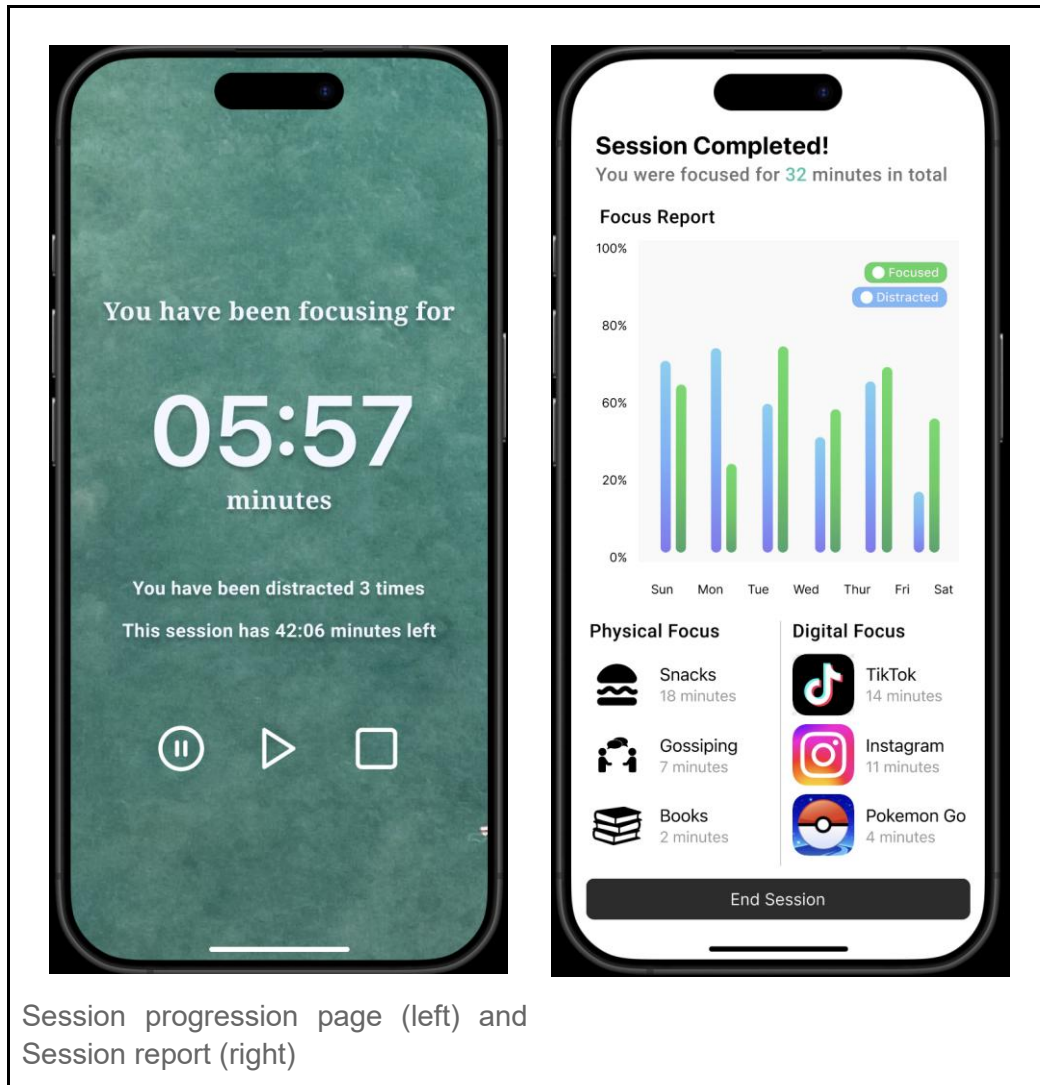
The key brand colour of our product is lake green. The colour blue implies professionalism, and green implies peace. By combining these two colours, as well as using white for contrast, we intend to inform the user that the app is meant for serious purposes, while not seeming as intimidating as perhaps some other clinical software. Our logo is designed to look like a monk wearing a shawl in order to perpetuate our message of inner focus, with the number 1 being subtly made using the body of the monk.



(From left to right: 1. Home page, 2. Comparison with friends, and 3. User forums)

After logging in, the user is then presented with the landing page of this app, which displays a summary of their progress this month. This creates intrinsic motivation for the user as they are able to see their records go up over time. Below is a quick select menu from which the user can select the mode of their focus period, as well as its duration. Unlike other similar apps, the options for the focus session are not stashed away under a second menu but instead presented as soon as the app is opened, as we want users to feel that our app is very efficient.

On the tab bar are four buttons, the first of which is the home page described above. The second is a "Friends" button, from which the user is able to compare how focused they've been with the friends they have added. Inspired by the WeChat fitness leaderboard, this feature creates intrinsic motivation for the user to compete with their friends, leading to longer focus periods. The third button leads to a user forums page, which is useful as it allows our more driven users to discuss potential self-improvement techniques with other like-minded people, fostering a community around the app. Lastly is the user options menu, omitted here as there is nothing particularly notable about it compared to other apps.



By clicking on the “Begin Session” button (try this in the demo; a few of these buttons have been implemented), the user is taken to a plain screen indicating how long they’ve been focusing. Users can start and pause the timer and wait until the session time is over or manually end the session early with the square button. This page shows when the physical and digital monitoring start and when the device camera will be used. We want to make it very clear to the user that these are the only times when monitoring occurs and that all data is stored and processed locally to avoid any privacy concerns regarding personal data.

After the session has ended, the user will be able to see a feedback report for the session, and the data from this session will also be immediately reflected on the user’s homepage. The session report will also include what the digital distractions have been for the user (by detecting which apps have been used during the session) and what kind of physical distractions the user has been engaged with. While the feature for detecting the type of physical distraction engaged has not yet been implemented, we hope to eventually develop our app to do so with more sophisticated computer vision and processing algorithms.

Business Plan

Executive Summary

Written by: Sean Wicaksana

One by One aims to revolutionise productivity by providing personalised, real-time solutions that help individuals and organisations focus on efficiently accomplishing their goals. We provide a cutting-edge AI-powered productivity platform to address the widespread issue of disrupted focus in work and school spaces, helping users concentrate seamlessly every day.

The worldwide productivity tools market is expected to increase by 13–15% annually, reaching \$78.82 billion in 2024. This presents a business opportunity, which our product can address by tackling physical and digital distractions simultaneously. With a target audience of professionals, students, and organisations across North America, Europe, and Asia-Pacific, One by One stands assured of capitalising on this substantial market opportunity.

One by One integrates multiple real-time inputs—eye and head tracking, body posture analysis, and digital activity monitoring—alongside personalised productivity recommendations. Unlike competitors, One by One goes beyond supervision, offering tools such as AI-powered focus recommendations, Pomodoro timers, leaderboards, and in-depth productivity reports. Leveraging advanced AI/ML ensures a tailored user experience that adapts to individual distraction patterns and goals.

One by One employs a freemium subscription model, with our main revenue streams being premium subscriptions, in-app advertising, and B2B sales. Our pricing strategy ensures accessibility while maintaining a competitive edge. The app will be distributed via app stores, browser extensions, and corporate partnerships.

Our main promotional strategies are influencer marketing, targeted advertisements, and collaborations with key opinion leaders (KOLs) in productivity and education. We focus on trust-building and organic user acquisition, complemented by a robust app store presence and direct outreach to organisations.

Financially, we project exponential growth. We seek funding requirements of 70,000 HKD, sourced through grants, competitions, and contributions. Initial losses in the first year will be offset by our increasing user base, with projections showing a break-even within the first 12 months of operation.

The One by One team comprises skilled professionals specialising in software development, business strategy, and financial planning. Our technical team ensures top-tier system performance, while our business team focuses on market analysis and strategic growth.

Company Overview

Written by: Sean Wicaksana

The Nature of Our Business

One by One operates in the technology and productivity solutions industry, offering a cutting-edge platform to enhance individual and organisational efficiency. Our product is a tool that helps manage scheduling and supervise the execution of each task. The business mainly leverages AI and sensor-based technologies for real-time monitoring and personalised productivity recommendations.

Our product will be available on mobile, desktop, and as a browser extension, catering to personal and corporate needs. By blending advanced machine learning, intuitive design, and real-world usability, One by One serves the growing market demand for practical productivity management tools in personal, academic, and professional settings.

One by One's Vision

To make productivity a natural and effortless part of life, empower people to stay focused in a world of distractions.

One by One's Mission

- One by One delivers real-time, personalised guidance to help individuals stay focused and productive with gentle, effective interventions.
- One by One simplifies task management, enabling effortless progress toward goals.

One by One's Values

- Continuous improvement: We believe there's always room for improvement, and we're always ready to fill it.
- Resilience: We prioritise returning to the right path over stagnating in the wrong.
- Collaboration: One by One believes in working together as a team and aligning the interests of the users with our own.
- Empowerment: We dream big, and we make it happen. We help people expand their capabilities and achieve their potential by encouraging strong will and determination.

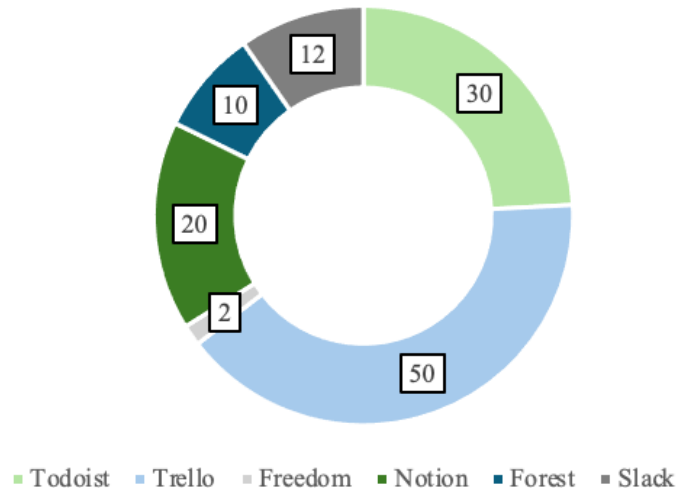
With One by One, Impressive is progressive.

Market Analysis

Written by: Lau Wai Man

Current Market Composition

Number of users(million)



Source from the official website of each app

From the above, we can see that Todoist, Trello, and Notion are the major competitors of our app. The total productivity app revenue in 2023 is \$1.43bn USD, with a total market size of \$78.82bn USD and an annual growth rate of around 13-15%

Market Trends

In the workplace, an average of 84.4% of employees are distracted during work (Yaqub, 2024). Currently, distractions from work cause significant economic loss for companies worldwide. In the US alone, workplace distraction causes an estimated economic loss to businesses of around \$650 bn USD. On average, employees lose 2.1 working hours per day due to distraction, which needs at least 25 minutes to return their focus to work.

In another area which needs anti-distraction, studying, 68% of students find they are bothered by smartphones, even though they understand that they should not be distracted (Nick, 2019). Hence, given that such a large percentage of the student population understands that they need some solution to their problem, they will be a key demographic for us in our marketing.

SWOT Analysis

Beneficial		Harmful
Internal	Strengths: Compared with our competitors, we provide the unique features of physical detection and employee comparison (for office workers).	Weaknesses: As we include the models of distance, position, and chat detection, which our major competitors do not, the system's price will be higher.
External	Opportunities: According to Neal Taparia (2023), the respondent's average distracted time is 42 minutes, while the actual distraction time is 2.7 hours. Hence, most respondents understand that they should not be distracted for long, which our app helps with.	Threats: According to B2B reviews (2024), 65% of workers want remote work, suggesting they want more freedom over their work. In this case, our apps may need to improve to fit the specific needs of people who don't follow a schedule, such as those with kids or pets.

PEST Analysis

<u>Political Factors</u> Considering the top 5 countries with the highest GDP, excluding India, all face issues with NEET (Not in employment, education, or training) people. Authorities adopted policies that help solve these issues, especially with China's anti "Tang Ping" movement. With One by One, we could help rebuild the attention of youth, helping them reconstruct a healthy lifestyle.	<u>Economic Factors</u> UN Trade and Development (2024) forecast that economic growth will slow to 2.6% in 2024, the third consecutive year slower than the pre-pandemic rate. Global income rates are lower than pre-pandemic when adjusted for inflation, meaning users will have less disposable income for our services.
<u>Social Factors</u> Society currently cares more about work-life balance than before. A report from Zippia found that 72% of employees found work-life balance essential for their daily life. Hence, we could use a focus app to help them finish their work efficiently and have more free time for later.	<u>Technological Factors</u> There are 5.170 billion active social media users globally in 2024, a 4.445% increase compared to 2023. Hence, this may lead to a more significant problem of being distracted by social media during work, going against the aim of our product.

Product and Operational Analysis

Written by: Sean Wicaksana

Product Description

One by One is an advanced comprehensive sensor-based system that integrates multiple real-time inputs and provides active real-time intervention to prevent distraction.

Multiple Real-time Inputs:

- **Overall Body Posture Changes**

One by One analyses the user's body language. Slouching, propping up the head on a hand, and other lost-focus nonverbal cues might indicate that the user is disengaged or distracted. Real-time analysis can signal when the users are losing focus.

- **Head and Eye Tracking Movements**

One by One, through the device's webcam, detects where you're looking to ensure you focus on your work. It can signal an interruption if your eyes stray from the task (e.g. head tilting more than 20 degrees to one side, being away more than 0,7 metres from the screen, and eyes away from the screen).

- **Comprehensive Digital Activity Supervision**

One by One could monitor mouse and keyboard activity. Lack of typing or excessive switching between apps could suggest distractions. Additionally, other app activities whose contexts are unrelated to work might indicate distractedness.

- **Objects-used detections**

One by One detects whether an object will likely support the user in reaching their goals.

- **Many more to come**

Once One by One determines absent-mindedness from these inputs, One by One will friendly remind you with a pop-up notification which contains either of the following:

- a. Encouraging motivational quotes
- b. Rest timer options
- c. Tips for taking adequate rest or water reminder
- d. Personalised user productivity feedback

Additionally, One by One provides users with several features to achieve peak productivity:

1. **Deadlines Calendar, To-do list, and Time blocking**
Plan tasks effectively with its integrated scheduling tools. One by One will also help adapt and adjust the timetables if unforeseen changes occur by its machine learning algorithm.
2. **Pomodoro timer**
One by One guarantees controlled work/rest intervals for intra- and inter-tasks to maintain the focus span throughout the working time.

3. **Productivity** **reports**
Hourly, daily, weekly, monthly, quarterly, and yearly statistics to track growth.
4. **Leaderboards**
Compare performance with global users if opted-in.

AI and Machine Learning Integration

One by One makes each productive experience personalised by using AI/ML to:

1. **Learn each user's distraction patterns**
By learning each user's habitual focus span, rest time, and frequent distraction forms, One by One can prevent distractions before they occur.
2. **Reduce the Rate of False Positives**
By analysing each user's background, habits, professions, necessities, and goals, One by One can have a proper foundation for determining the potential rate of particular physical objects and digital application activities to be seen as disturbance signs.
3. **Provide AI-powered focus recommendations**
With a complete understanding of the users, One by One can give personalised feedback, encouragement, and tips for taking rest effectively.
4. **Provide personalised behaviour analysis reports**
After each focus session, One by One will produce a brief report to inform each user of their primary areas of distraction and how to overcome them in the future.

Mode of Use

One by One has three settings to suit various users' needs:

- **Personal Mode:** This mode supports users in setting and achieving their personal goals, whether for work, study, or self-improvement.
- **School Mode:** This mode enables teachers to monitor and guide students' focus and progress, providing insights into their learning patterns.
- **Work Mode:** This remote work-friendly mode allows companies to track the overall productivity of their teams. One by One also supports filling in timesheets or timetable reports for their managers.

Unique Selling Points

Compared to other apps in the market, One by One has several competitive advantages:

1. One by One combines multiple real-time inputs beyond digital activities.
2. One by One embeds cutting-edge AI/ML technology for personalisation by learning each individual's distraction patterns and tailoring interventions accordingly.
3. Instead of just supervising, we also guide users step by step using a complete suite of productivity tools.
4. One by One is suitable for personal use, team settings, or corporate environments, making it adaptable to various professions and industries.

5. One by One, in a non-intrusive and focus-enhancing manner, proactively anticipates distractions instead of just reacting to them.

Distribution Plan

As One by One will be accessible in the form of Apps and Web Extensions, One by One will mainly be distributed via the following channels:

1. Apps and Web Extensions Marketplaces

One by One will be available on major platforms like Google Play Store, Apple App Store, Microsoft Store, etc., and browser extension marketplaces.

2. Corporate and Organisations Partnerships
Direct sales to businesses as bundled packages for team productivity management.

Financial Analysis

Written by: Li Zhifang

Costs and Revenues of Production, Distribution, and Marketing

While the costs of making software may be as high as a couple thousand USD¹, we as a team can produce the software at nearly no cost, given that we currently possess, or are willing to learn, the skills required to publish software from scratch. Our expenses are thus limited to distribution and marketing, aside from the occasional employment of specific expertise, such as advice on streamlining our system architecture.

As seen by our marketing strategy, a key publicity component will be collaborations with social media influence and Key Opinion Leaders (KOL). Due to the funding limitations, we are not aiming for expensive sponsorships such as product reviews but relatively pure outreach to potentially interested influencers, costing time rather than money. Instead, we will use Google Ads for cheap yet effectively targeted advertisements.

The user base, and thus associated costs, of this app can be modelled by adding an exponential to a logarithmic curve. This is because we will begin with very few users and thus boost the software with paid traffic in the form of Google Ads. Over time, however, more of our users will have discovered us organically, thus helping our product have a higher profit margin.

$$\text{User Count } (u) = a + b, \text{Organic users } (a) = \frac{1}{10} \times e^{3+x}, \text{Acquired users } (b) = 100 \times \ln(3 + x)$$

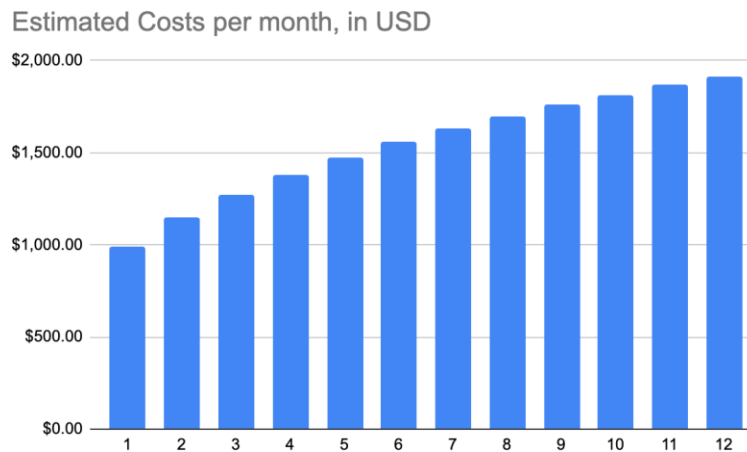
¹ Artem Dogtiev, "App Development Cost (2024)," Business of Apps, September 23, 2024, <https://www.businessofapps.com/app-developers/research/app-development-cost>.

These equations follow the previous model, and the parameters are found to best match the average for new software².

Assuming a conversion rate of 5% and using Google's ad cost of \$0.35 per click, the average customer acquisition cost (CAC) is \$7 USD³. Hence,

$$\text{Cost of Acquiring Customers} = 7 \times [100 \times \ln(3 + x)]$$

Adding on 20 USD per month in server costs, we can make the following projections,



Given costs, we can calculate net profit by considering sales revenue. Our user base is split into two groups: paying and non-paying customers. The estimated 5% of paying customers⁴ receive no ads in exchange for \$5 USD per month. For the remaining 95%, we can use the Google AdSense model of \$1 USD per 1000 views⁵. Assuming each of our active users watches two ads each day, $\text{Sales Revenue} = 0.95 u \times 2 \div 1000 + 0.05 u \times 5$. Thus, by our current values, we can project net profit as follows.

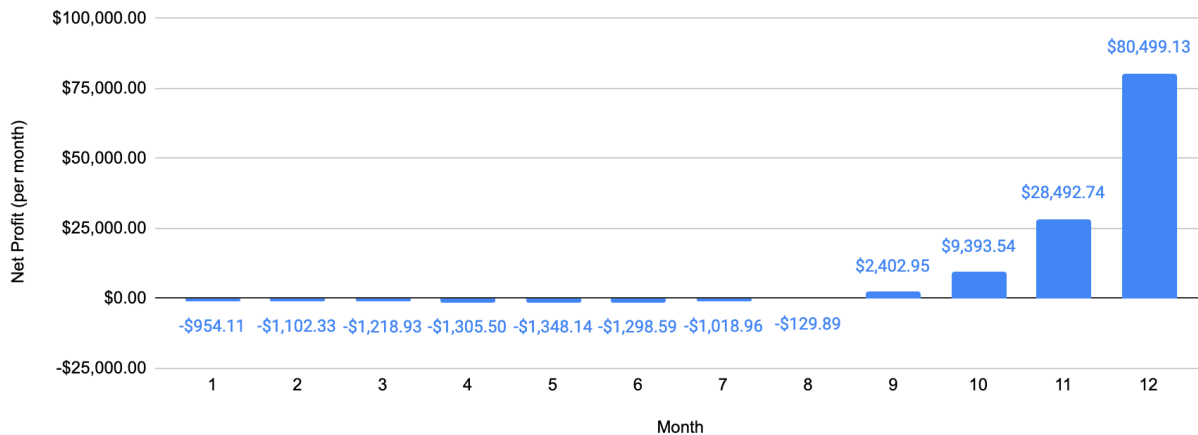
² Mark Wilcox, "How Many Users Is Realistic?," Developer Nation, February 22, 2013, <https://www.developernation.net/blog/how-many-users-is-realistic/>.

³ Abby Fields, "How Much Does Google Ads Cost in 2023?," WebFX, August 14, 2024, <https://www.webfx.com/blog/ppc/much-cost-advertise-google-adwords/>.

⁴ Khalid Saleh, "Global In-App Purchase Revenue Statistics and Trends [Infographic]," The Invesp Blog, November 1, 2017, <https://www.invespcro.com/blog/in-app-purchase-revenue/>.

⁵ Ben Rycroft, "AdSense Revenue Calculator Tool: Estimate Your Earnings (2023)," Snigel, January 4, 2024, <https://snigel.com/blog/adsense-revenue-calculator>.

Net Profit of app (in USD per month)

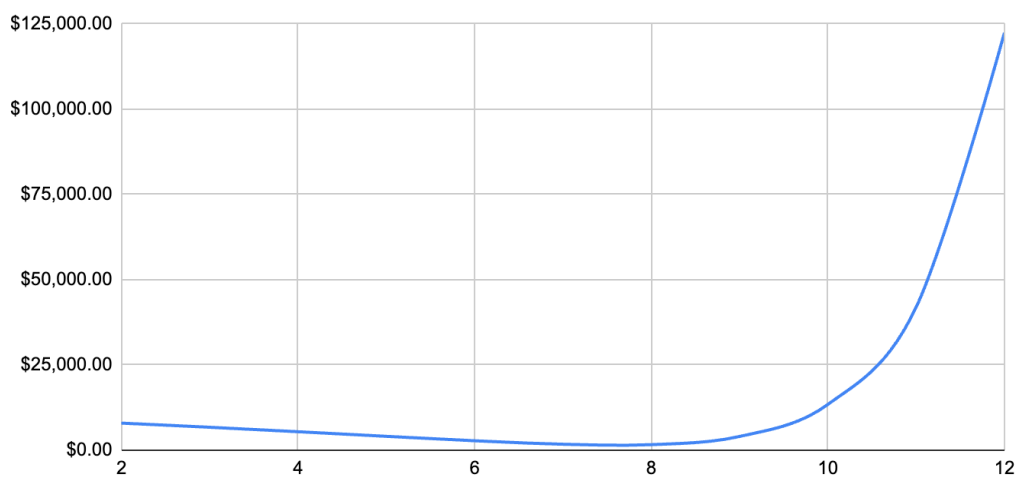


Budget and Funding

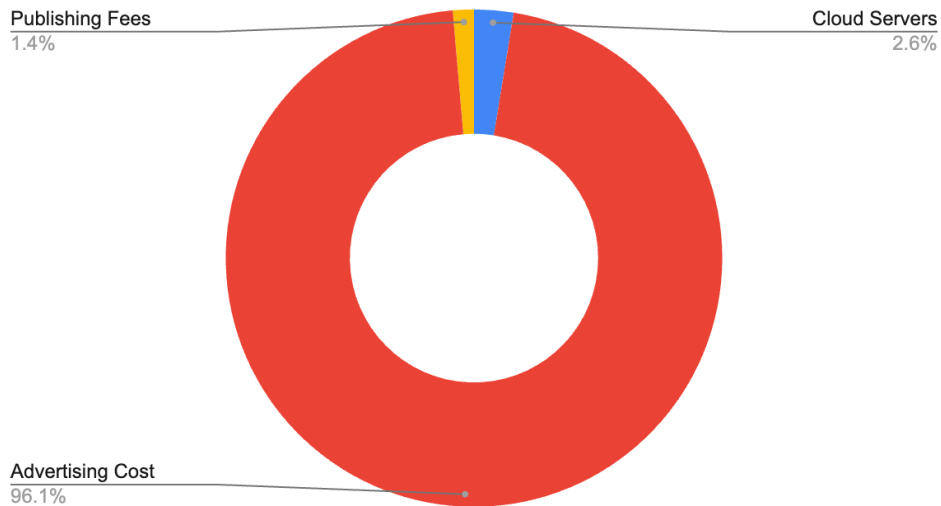
We aim to raise 70,000 HKD through university grants and funding and the various competitions the HKU Techno-Entrepreneurship core offers each year. As a team, we are fortunate enough to be in an environment where plenty of funding is available. If internal funding is not possible and investors cannot be found, we will seek funding outside the university, possibly from subsidies the Hong Kong government provides to Science Park.

For the rest of the funds, we're each willing to put 2,000 HKD of our savings into the project. All of this is needed as our projections show that we will be negative in profit for the first 8 months and will thus need some cash on hand to keep us afloat, alongside being conservative with our grant estimations. By using this fund money, we will be able to continue operating this business until it is able to generate exponentially more money.

Cash balance each month, in USD (with 10,000 USD starting cash)



Budget Distribution According to Cost



As can be seen, most of our funds will be spent on advertising, to which we will contribute all of our grant money. This is because our product can succeed by itself, and we are using the funds solely to boost the product to grow further. All other expenses, including programming and design, will be covered by our group directly, and we will not be taking pay from the app until it is entirely self-sufficient.

Cash Flow Forecast

Below is the cash flow forecast for the first year using the aforementioned formulae

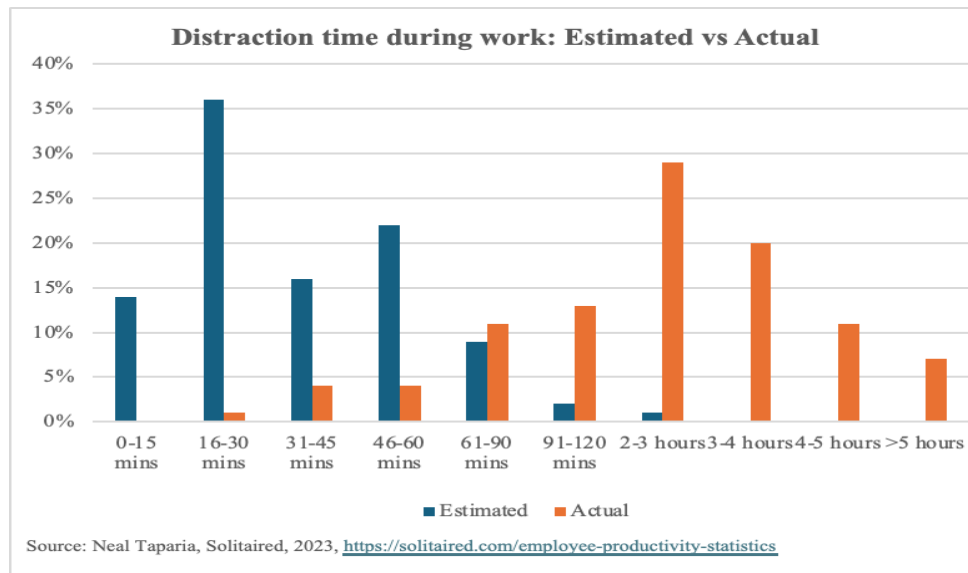
Month	User estimate	Advertising Cost (USD)	Sales Revenue (USD)	Server Cost (USD)	Monthly Profit (USD)	Cash Balance (USD)
1	144	\$970.41	\$36.30	\$20.00	-\$954.11	\$9,045.89
2	176	\$1,126.61	\$44.28	\$20.00	-\$1,102.33	\$7,943.56
3	220	\$1,254.23	\$55.30	\$20.00	-\$1,218.93	\$6,724.63
4	304	\$1,362.14	\$76.64	\$20.00	-\$1,305.50	\$5,419.13
5	506	\$1,455.61	\$127.47	\$20.00	-\$1,348.14	\$4,071.00
6	1,030	\$1,538.06	\$259.46	\$20.00	-\$1,298.59	\$2,772.40
7	2,433	\$1,611.81	\$612.85	\$20.00	-\$1,018.96	\$1,753.44
8	6,227	\$1,678.53	\$1,568.63	\$20.00	-\$129.89	\$1,623.55
9	16,524	\$1,739.43	\$4,162.39	\$20.00	\$2,402.95	\$4,026.50
10	44,498	\$1,795.46	\$11,209.00	\$20.00	\$9,393.54	\$13,420.04
11	120,524	\$1,847.34	\$30,360.08	\$20.00	\$28,492.74	\$41,912.78
12	327,173	\$1,895.64	\$82,414.76	\$20.00	\$80,499.13	\$122,411.91

As with every business, the initial revenue is low, and the burden falls on our shoulders. Still, it will become exponentially more profitable over time, especially past the first year of operation.

Marketing Strategy

Written by: Lau Wai Man

Target Demographics

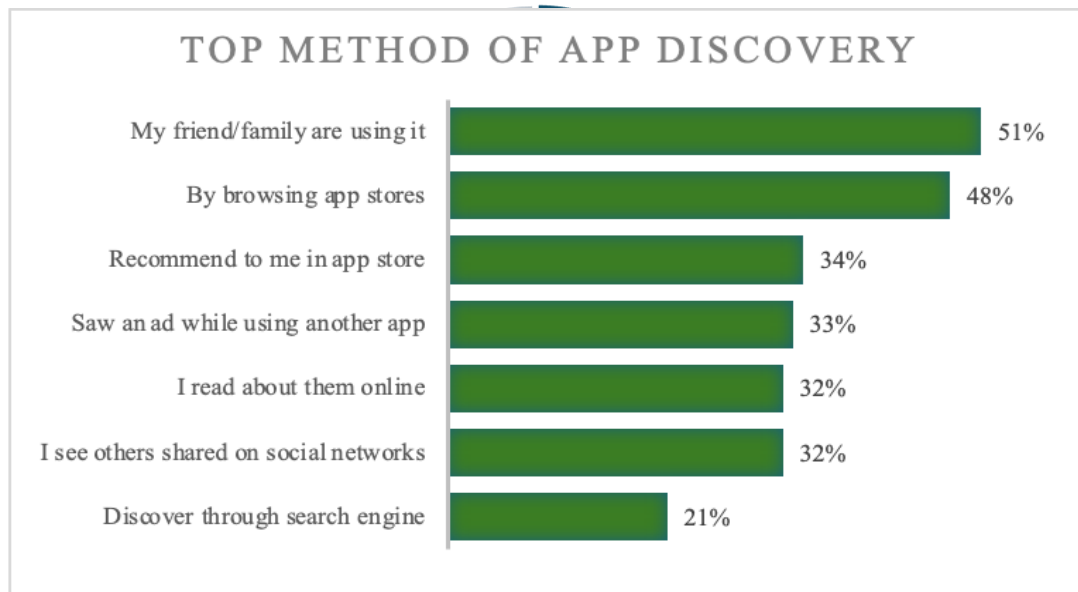


A report studying the distraction hours during work shows that 84.4% of employees are distracted at work, and the actual amount was almost four times the estimate. Given the number of distracted workers, we can sell this product both to companies seeking to maximise their efficiency and to very motivated workers. Students and parents are also a target demographic as they are similarly found to be very often distracted.

Market Segmentation

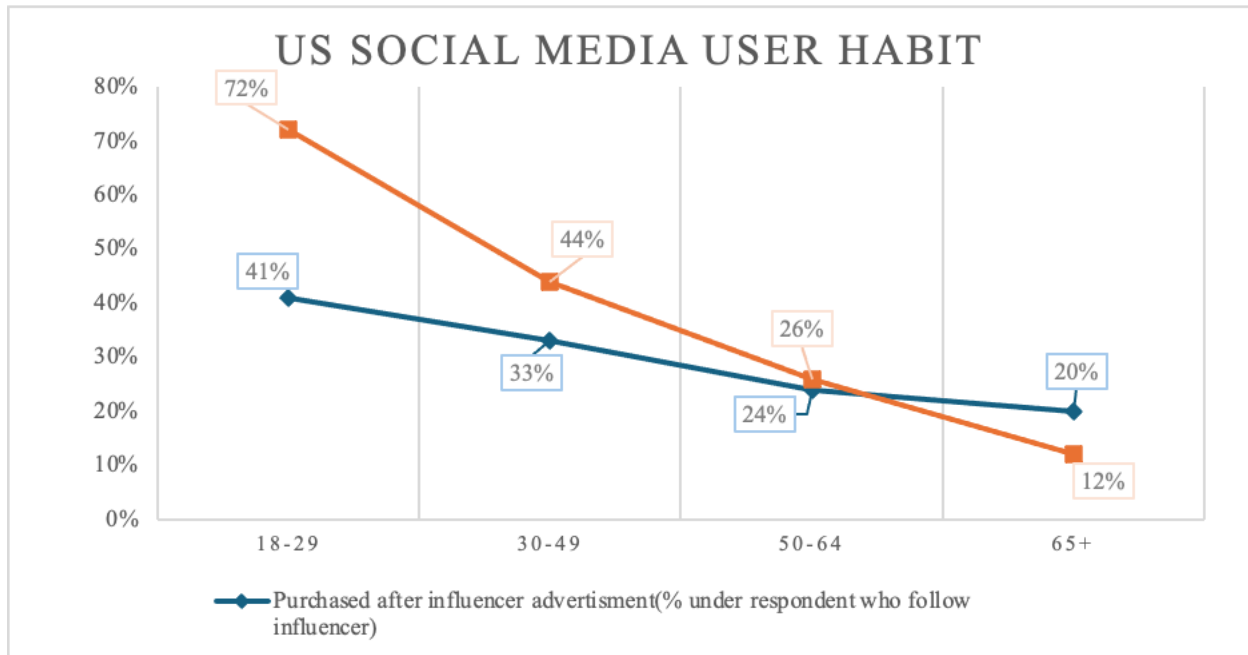
From the diagram below, our major targeted regions will be North America, Europe, and Asia Pacific, including most developed countries.

Market Revenue of Productivity App



Promotional Strategies

The table shows that some of the top methods of new app discovery for internet users include browsing app stores and recommendations from others. According to Matter findings (as cited in Deyo, 2023), 69% of consumers trust influencers, friends and family over information coming directly from a brand.



From the above table, social media influencers could change the decisions of social media users, especially for users between the ages of 18 and 49. Hence, our group believed that according to the above analysis of internet users' habits, we could use the influence of key opinion leaders (KOL) on social media as our promotion strategy.

Competitor Analysis

Our research on other existing productivity and focus software shows that our major competitors are Freedom, Cold Turkey Blocker, and the screen time function on your phone. We find that other than Freedom, all the other major competitors are completely free of charge. At the same time, however, their functions are also limited to detecting online activities such as blocking apps and websites. According to Malyshev (2024), daydreaming and chatting with colleagues were at the top of the list of activities that distracted employees, which our app is able to detect through its distance and posture detection, filling that gap in the market.

Our competitors mainly used influencers to promote their products. The KOLs (bloggers, YouTubers) that promoted these products were primarily in the technology sectors, which helped give their opinions more credibility. Considering this, we will narrow our scope to specific influencers that our audience can relate to for promotion (e.g. KOLs with content that mainly focuses on their daily lives or sharing study strategies). The daily usage of this app by KOLs would help showcase its effectiveness to our key demographics.

Pricing Strategies

Plan (yearly)	Cost (USD) per year
Adobe (with 7-days free trial)	\$179.88
Freedom	\$39.9
AppBlock	\$29.99
Microsoft (individual)	\$150

Source from official websites of each app

According to the table above, which are some apps that would be used in the workplace, the price range of focus apps is between \$19.99 - 39.99 USD. The productivity apps mentioned also have promotion plans that could attract users through discounts or free trials. Considering the app's selling point is its effectiveness in avoiding distraction, we believe that it is necessary for consumers to have a trial before subscribing to the plans as they will immediately experience its benefits. Hence, we can use a free model for basic features and add distance detection, chat detection, and forum features as part of our premium plan. Knowing that such focus apps may need a short period of time before the users fully experience their effectiveness, we plan to have a 15-day free trial for different types of users. Besides that, as One by One was still a newly established app, we believe the price could be slightly lower than the average level to attract more users. Therefore, by considering all of the above, the pricing strategies of our app will have a 15-day free trial with a monthly price of \$2.99 USD in the first three months and \$5 USD afterwards.

Main Challenges and Solutions

Core Contributors: Wong Suet Nga / Zeng Xinyu / Chau Wai Yee / ZhangJiachang

1. Data Privacy and Security

Data privacy and security are key challenges for us both in terms of ethics and marketing, as our users may be concerned that sensitive personal data is being secretly stored and sold to third parties. This particularly concerns the chat monitoring and facial recognition aspects of our app. As such, all of our data is cached and processed locally, being deleted as soon as each session ends. Additionally, we are implementing a strict user agreement regarding our analytical functions, from which our customers can withdraw at any time. As an ethical company, our agreement states that all videos and images taken by the app are used for analytical purposes only and will under no circumstances be shared with a third party. Additionally, all of our user data will be encrypted as we understand the danger of possible malicious attacks, and we plan to conduct regular security assessments after the software's launch.

2. Identification of the User Object in Distance Detection

There are two main challenges when it comes to distance detection. First, we need to identify that the user is the object used for distance detection amongst all other objects in the frame, all of which are at an unknown distance away from the camera. Secondly, we need to reduce computational load, as we want the software to be lightweight, and the processing might be slow.

It was found that trade_id 1 is the ID for the user object. Before calculating the distance, we would check if the trade_id is equal to 1 so that we can find the user object among all detected objects to calculate the distance only for the user object and reduce computational load.

3. Garbled Text in Chat Analysis

In the chat analysis implementation, the text extracted is often garbled. This is likely due to the fact that pyTesseract extracts text from an image horizontally, line by line. To illustrate, take an example from the following screenshot:



Ideally, the words will be read as “Please read the latest run down and arrangements”, when in reality, the words extracted are “**ri** robocon group i **vvtue youq[ephot d}muf p }wk** please read the latest run down and **arrangements17:45 ii*hr&seopnq**” - clearly a lot of garbage text. This primarily results from graphics and the incorrect linkage of characters, with the effect greatly amplified when images and emojis are involved. Some text that may have been meaningful (such as “arrangements”) will become “meaningless” when characters are incorrectly linked together (e.g., arrangements17:45).

4. Cross-Platform Challenges

The development environments for Android, iOS, and web browsers differ significantly, which can cause problems for our software. Our team uses Python as our main programming language, and it is difficult to unify all of our functions under a single Python script. Differences in operating systems and version differences can also lead to unexpected errors, all of which will increase our maintenance costs. To reduce this, we plan to transition to a cross-platform framework such as Kivy, and common modules will be written for reuse whenever possible.

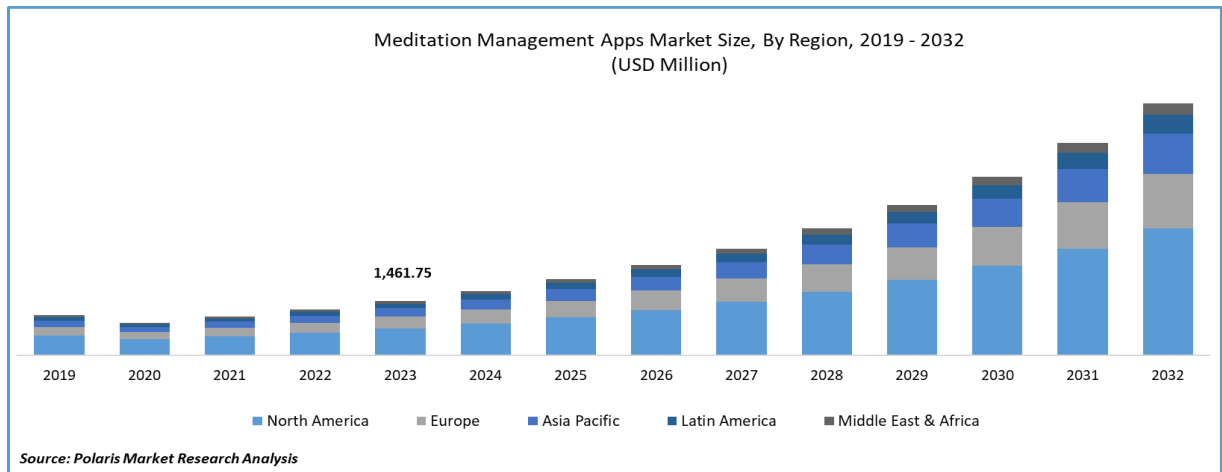
Possible Future Developments

Written by: Lau Wai Man

Employees nowadays are facing lots of pressure due to various forces in the workplace, leading to an increase in reported mental stress. As stated by a report from AXA in 2024, 77% of workers have experienced at least one mental health issue due to their jobs, which is a significant increase compared to 27% in 2018 (CMHA, 2020). Knowing that mental health is one of the vital challenges

faced by our potential customers, we believe that our users' mental health is one of the key areas worth investing in when it comes to future development.

From the graph above, we can see that the market size for Meditation Management Apps will have an expected growth in the next 8 years and an estimated CAGR of 18.1% between 2024 and 2032. Hence, there is vast potential in adding mindfulness functions into our app, which follows both the self-improvement mission of our app and targets this rapidly growing market.



Additional Notes

Video credits go to Wong Suet Nga and Sean Wicaksana Faustine

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